TANMOY ROY *PhD*

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SUMMARY

Senior Data Scientist and **PhD** with 19+ years delivering AI/ML systems in healthcare, finance, and telecom. Currently at DeliverHealth, leading on-prem **small language model** (SLM) solutioning: medical coding co-pilots, automatic clinical notes, RAG correction, and medical ASR. Collaborates with product and partners including Google and Microsoft. Focus: *STT, RAG, SLM fine-tuning, clinical NLP, computer vision*.

Career started at **SIEMENS** (8+ years) building telecom, electronic-vehicle, and smart-grid billing systems. Mid-career PhD and post-doctoral fellowship at the **University of Johannesburg** (2016–2021). Doctoral research was **speech emotion recognition** (SER): a novel feature set (SEGRT) and a speech endpoint-detection algorithm, with peer-reviewed publications. Then returned to applied AI/ML.

EXPERIENCE

DeliverHealth Solutions

Remote, India [July '24 – Present]

Senior Data Scientist

Lead AI/ML product ideation and organization-wide adoption. Design and deploy on-prem specialist models with product owners and partners including Google and Microsoft. Current bar: models small enough to inspect and honest enough to defer, not merely more fluent.

- **Medical Coding Co-pilot:** SLM fine-tuned on proprietary historical data in a continuous learning loop to assist human coders and cut time-per-document. White-box logits used as a confidence meter; on this task a specialized on-prem SLM outperformed a generic LLM (Gemini) baseline, with right-vs-wrong AUROC moving from $\sim$0.60 to $\sim$0.90.
- **Automatic Medical Note Generation:** Template-based physician note generation on a self-service platform, using Gemini 3.1 Pro and proprietary historical data.
- **RAG-based Medical Correction Engine:** Retrieval-augmented correction of ASR drafts before delivery, reducing manual cleanup and lifting document quality.
- **Medical ASR (Med-ASR):** In-house speech-to-text engine on proprietary data. Fine-tuned Gemini-1.5-Pro and Whisper; reached 12% medical WER without sending encounter audio to a public cloud.

Cloudcraftz Solutions

Kolkata, India [Nov '22 – Jun '24]

Senior Data Scientist

Led and mentored a team spanning data science, quantitative analytics, and software engineering, delivering finance-domain systems.

- **Option Trading Strategy Backtest:** Production backtest for complex option strategies.
- **Explainable AI:** SHAP and accumulated local effect (ALE) visualizations for model transparency; built the product end to end.
- **Data Visualization Platform:** Low-intervention analytics UI; led platform development.

Veersa Technologies

Kolkata, India [May '22 – Oct '22]

Data Scientist (Consulting)

Primary responsibility was ML model building with both tabular and image data. Other responsibilities involved data engineering, ML model productionizing, and ML team building. Executed projects in the healthcare domain.

- **Insurance Claim Denial Prediction:** Predict whether claims submitted by the health institute will get denied or approved, and show the reasons behind the denial. Tabular data with relevant features, after pre-processing, was the training data for an XGBoost classifier. Model accuracy was 89% although there was further scope for improvement.
- **Patient Data De-Duplication:** A broken chain of information affects service quality for healthcare providers and consumes storage space. Used the record linkage concept to solve this problem.
- **Disease Detection from X-Ray images:** Used a ResNet34 model to achieve 84% accuracy.
- **Clinical Trial Enrollment Prediction:** Proof of concept on publicly available high-quality data. The target was to predict the number of people finally enrolled for a clinical trial. Prediction accuracy was ∼75% but there was further scope for improvement.
- **Handwritten Medical Form Reader:** Varying handwriting styles make these forms hard to digitize. Used AWS Textract to extract the handwritten text and built the solution around the Textract output.

Research and Training Consultancy

Kolkata, India [Feb '20 – Present]

Research Consultant

Research consultancy across computer vision, NLP, and data science:

- **Human Fall Detection (CV):** Fall detection from surveillance frames. Evaluated current deep-learning modules. Proposed a Masked R-CNN semantic segmentation model [8] at 94% F1 and 89% IoU; next module is human pose estimation.
- **Text Classification (NLP):** Multi-head attention classifier [1] above 96% test accuracy, compared with Naive Bayes, logistic regression, random forest, and MLP.
- **Polyp Detection (Medical Imaging):** U-Net architecture for polyps in colonoscopy images, aimed at cancerous-polyp detection.
- **Electronic Health Record (EHR) and Process Mining:** 30-day readmission prediction from EHR. Process-mining features into sequence-to-sequence models; 88% test accuracy.
- **Face Mask Detection (CV):** Mask detection from surveillance cameras during Covid-19 using MobileNet V2 and Inception ResNetV2.

Training Consultant

Training tracks for students, career-switchers, and practising engineers. Topics: Data Science & ML, Deep Learning (ANN/CNN/RNN/Attention), Python for ML/DL, Practical Machine Learning (MLOps), and Cloud & Docker (AWS DMS, RDS, EMR, SageMaker).

University of Johannesburg

Johannesburg, South Africa [Jul '16 – Sep '21]

Post Doctoral Research Fellow [Oct '19 – Sep '21]

Built a feed-forward network [2] on the doctoral SER feature set [7]. Wrote two surveys: emotion recognition in the 4th industrial revolution [4] and SER classification techniques [3]. Opened collaborations with King Saud University [8] and BITS Pilani.

Doctoral Research Fellow [Jul '16 – Jun '19]

Thesis: "Detecting emotions from speech using Machine Learning Techniques," completed in three years. Proposed SEGRT (*subjective emotional gap reduction technique*) [7, 5] and a speech endpoint-detection algorithm for pre-processing [6].

SIEMENS

Kolkata, India [Jan '07 – Sep '15]

Associate Consultant [Jan '13 – Sep '15]

• Designed and implemented a Telecom Service Provisioning system based on a RedHat Linux multi-tiered application using Java Spring Framework. Technologies used *Java Spring, Oracle*.
• Customization of Telecom Service Provisioning system using *Java, SOAP, XML, Oracle* on *Linux* platform.
• Designed and implemented CRM Service Provisioning integration module for GSM-R Billing system using *Java, Oracle PL/SQL*.
• Designed and implemented the Billing module for GSM-R Billing system using *C/C++, Oracle PL/SQL* on *Linux* platform.

Sr. System Engineer [Jan '11 – Dec '12]

• Designed and implemented a Charging module for the Electronic Car billing system using *C/C++ and Oracle*.
• Designed and implemented GUI based Telecom Call Analysis system. Features include Call tracking, Zone based call analysis with Graphical simulations. Used *C++ based Qt* GUI Library.

System Engineer [Jan '09 – Dec '10]

• Designed and implemented Bill Printing and Bill By Email module for a Telecom Billing system. Technologies: Java, ProC/C++, Oracle PL/SQL.
• Designed and implemented a Charging module for a Telecom Interconnect Billing system. Technologies: Java, Oracle.

Associate System Engineer [Jan '07 – Dec '08]

• Single handedly provided System Support for IVR and USSD Solution. Technologies used: *Java, C++, Linux, awk script, shell script*.
• Developed an automated monitoring system using *Java, shell script, awk script* to improve client experience.

RESEARCH SUPERVISION & GUIDANCE

- Guided a PhD scholar from King Saud University, Saudi Arabia (completed). Computer Vision: detection of human fall from a surveillance camera.
- Supervised an MTech student from BITS Pilani, India (completed). Computer Vision: visual scene understanding for autonomous vehicles.

TECHNICAL EXPERTISE

- **Programming Languages:** Python, C/C++, R, Matlab, Java, JavaScript, SQL, Scala, Julia.
- **Libraries:** ollama, langchain, langgraph, unsloth, pytorch, tensorflow, pandas, matplotlib, scipy, opencv, pyspark.
- **Technologies/Platforms:** On-prem GPU inference, Cloud (AWS, GCP), Linux, Docker, MLOps (KubeFlow), REST, SOAP, XML.
- **Databases:** RDBMS (Oracle, PostgreSQL, SQL Server, MySQL), NoSQL (MongoDB, DynamoDB, Cassandra), Cloud DB (Amazon RDS).

EDUCATION

Doctor of Philosophy – University of Johannesburg, 2019. Applied ML & Speech Signal Processing.
Master of Science – Lovely Professional University, 2013. Information Technology.
PG Diploma in Computer Applications – DOEACC Society, 2004.
Bachelor of Science – University of Calcutta, 2002. Economics (Hons), CS, Mathematics.

SCHOLARSHIPS, REVIEWING AND CERTIFICATES

- Global Excellence Stature (GES) Doctoral Fellowship, University of Johannesburg, 2016.
- GES Post Doctoral Fellowship, University of Johannesburg, 2019.
- Journal Reviewer, *International Journal of Systems Science* (Taylor and Francis).
- Coursera: Deep Learning with PyTorch: Neural Style Transfer; Neural Network from Scratch in TensorFlow.

PUBLICATIONS

[1] Sinha J and Roy T. "Attention Based Encoder Architecture for Automatic Text Classification: A Case Study on Text-news Classification". In: *International Journal of Scientific Research in Engineering And Management* (2022). DOI: 10.55041/IJSREM12107.

[2] Roy T, Marwala T, and Chakraverty S. "Chapter 12 - Speech emotion recognition using deep learning". In: *New Paradigms in Computational Modeling and Its Applications*. Ed. by Snehashish Chakraverty. Academic Press, 2021, pp. 177–187. ISBN: 978-0-12-822133-4. DOI: https://doi.org/10.1016/B978-0-12-822133-4.00009-8.

[3] Roy T, Marwala T, and Chakraverty S. "A Survey of Classification Techniques in Speech Emotion Recognition". In: *Mathematical Methods in Interdisciplinary Sciences*. John Wiley & Sons, Ltd, 2020. Chap. 3, pp. 33–48. ISBN: 9781119585640. DOI: https://doi.org/10.1002/9781119585640.ch3. eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1002/9781119585640.ch3.

[4] Roy T, Marwala T, and Chakraverty S. "Advancements and Role of Emotion Recognition in the 4th Industrial Revolution". In: *The Disruptive Fourth Industrial Revolution: Technology, Society and Beyond*. Ed. by Wesley Doorsamy, Babu Sena Paul, and Tshilidzi Marwala. Cham: Springer International Publishing, 2020, pp. 179–203. ISBN: 978-3-030-48230-5. DOI: 10.1007/978-3-030-48230-5_8.

[5] Roy T, Marwala T, and Chakraverty S. "Speech Emotion Recognition Using Neural Network and Wavelet Features". In: *Recent Trends in Wave Mechanics and Vibrations*. Ed. by S Chakraverty and Paritosh Biswas. Singapore: Springer Singapore, 2020, pp. 427–438. ISBN: 978-981-15-0287-3. DOI: 10.1007/978-981-15-0287-3_30.

[6] Roy T, Chakraverty S, and Marwala T. "Precise detection of speech endpoints dynamically: A wavelet convolution based approach". In: *Communications in Nonlinear Science and Numerical Simulation* 67 (2019), pp. 162–175. ISSN: 1007-5704. DOI: https://doi.org/10.1016/j.cnsns.2018.07.008.

[7] Roy T et al. "Introducing New Feature Set based on Wavelets for Speech Emotion Classification". In: *2018 IEEE Applied Signal Processing Conference (ASPCON)*. 2018, pp. 124–128. DOI: 10.1109/ASPCON.2018.8748666.

In preparation

[8] Alanazi T and Roy T. "Generating Human Segmentation Dynamically in Multi-camera Setup Using Deep Learning". In preparation. 2022.